

PROBLEM No 1: (1+1+1 =3 points)

Determine the probability that an energy level is empty of an electron if the state is below the Fermi level by (a) kT , (b) $5 kT$, and (c) $10 kT$.

Solution:

$$f(E)_{Occupied} = \left[1 + e^{\left(\frac{E-E_F}{kT}\right)} \right]^{-1} \rightarrow f(E)_{Empty} = \left[1 + e^{\left(\frac{E_F-E}{kT}\right)} \right]^{-1}$$

$$a. \quad f(E_F - kT) = \left[1 + e^{\left(\frac{E_F+kT-E_F}{kT}\right)} \right]^{-1} = \left[1 + e^{\left(\frac{kT}{kT}\right)} \right]^{-1} = [1 + e^{(1)}]^{-1} = 0.26894142137$$

$$b. \quad f(E_F - 5kT) = \left[1 + e^{\left(\frac{E_F+5kT-E_F}{kT}\right)} \right]^{-1} = \left[1 + e^{\left(\frac{5kT}{kT}\right)} \right]^{-1} = [1 + e^{(5)}]^{-1} = 0.0066928509243$$

$$c. \quad f(E_F - 10kT) = \left[1 + e^{\left(\frac{E_F+10kT-E_F}{kT}\right)} \right]^{-1} = \left[1 + e^{\left(\frac{10kT}{kT}\right)} \right]^{-1} = [1 + e^{(10)}]^{-1} = 4.5397868702434 * 10^{-5}$$

PROBLEM No. 2 (2 points)

The probability that a state at $E_c + kT$ is occupied by an electron is equal to the probability that a state at $E_v - kT$ is empty. Determine the position of the Fermi energy level as a function of E_c and E_v .

Solution:

$$f(E)_{Occupied} = \left[1 + e^{\left(\frac{E-E_F}{kT}\right)} \right]^{-1} \rightarrow f(E)_{Empty} = \left[1 + e^{\left(\frac{E_F-E}{kT}\right)} \right]^{-1}$$

$$f(E_c + kT) = \left[1 + e^{\left(\frac{E-E_F}{kT}\right)} \right]^{-1} = \left[1 + e^{\left(\frac{E_c+kT-E_F}{kT}\right)} \right]^{-1}$$

$$1 - f(E_v - kT) = \left[1 + e^{\left(\frac{E_F-E}{kT}\right)} \right]^{-1} = \left[1 + e^{\left(\frac{E_F-E_v+kT}{kT}\right)} \right]^{-1}$$

$$\left(1 + e^{\left(\frac{E_c+kT-E_F}{kT}\right)} \right)^{-1} = \left(1 + e^{\left(\frac{E_F-E_v+kT}{kT}\right)} \right)^{-1}$$

$$e^{\left(\frac{E_c+kT-E_F}{kT}\right)} = e^{\left(\frac{E_F-E_v+kT}{kT}\right)}$$

$$\frac{E_c + kT - E_F}{kT} = \frac{E_F - E_v + kT}{kT}$$

$$E_c + kT - E_F = E_F - E_v + kT$$

$$E_c - E_F = E_F - E_v$$

$$2E_F = (E_c + E_v)$$

$$E_F = \frac{E_c + E_v}{2}$$

Student's Name: Jacob Vaught

PROBLEM No 3: (2.5 + 2.5 + 2 = 7) points

- (a) Calculate the thermal equilibrium electron and hole concentration in silicon at $T = 350\text{K}$ for the case when Fermi Energy is 0.22 eV below the conduction band. Given that $E_g = 1.12\text{ eV}$; at 350K $N_c = 2.8 \times 10^{19}\text{ cm}^{-3}$, and $N_v = 1.05 \times 10^{19}\text{ cm}^{-3}$.
- (b) Draw a representative energy band diagram (using parallel lines) showing conduction band, valence band, Fermi energy level and intrinsic energy level. Mark band-gap and other energy levels.

Solution:

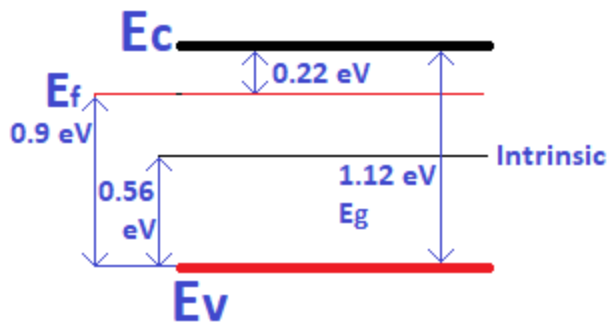
a.

Fermi energy is $(1.12 - 0.22 = 0.9\text{ eV})$ above the Valence Band.

$$n_0 = N_c * \exp\left(-\frac{E_c - E_F}{kT}\right) \rightarrow n_0 = 2.8 * 10^{19} * e^{\left(-\frac{0.22}{8.617 * 10^{-5} * 350}\right)} = 1.902 * 10^{16}$$

$$p_0 = N_v * \exp\left(-\frac{E_F - E_v}{kT}\right) = 1.05 * 10^{19} * e^{\left(-\frac{0.9}{8.617 * 10^{-5} * 350}\right)} = 1.15 * 10^6$$

b.



Student's Name: Jacob Vaught

PROBLEM No 4: (2 + 2 + 2 + 2 = 8) points

Given the effective density of states of holes, electrons, and the bandgaps in the in table provided at the end of this HW, calculate the position of Fermi levels with respect to center of the bandgap at 300K for

- Si
- Ge
- GaAs
- Plot all of the above (separately for each material) calculations using straight lines indicating conduction/valance bands mid-gap and Fermi levels.

Solution:

a. $N_c = 2.8 * 10^{19}$; $N_v = 1.04 * 10^{19}$; $E_v = 1.12$; $m_p^* = 0.56$; $m_n^* = 1.08$; $E_{Fi} - E_{midgap} = \frac{3}{4} kT * \ln\left(\frac{m_p^*}{m_n^*}\right)$

$$E_{Fi} - E_{midgap} = \frac{3}{4} * 8.617 * 10^{-5} * 300 * \ln\left(\frac{0.56}{1.08}\right) = -0.01273eV$$

b. $m_p^* = 0.37$; $m_n^* = 0.55$; $E_{Fi} - E_{midgap} = \frac{3}{4} kT * \ln\left(\frac{m_p^*}{m_n^*}\right)$

$$E_{Fi} - E_{midgap} = \frac{3}{4} * 8.617 * 10^{-5} * 300 * \ln\left(\frac{0.37}{0.55}\right) = -0.00768579eV$$

c. $m_p^* = 0.48$; $m_n^* = 0.067$; $E_{Fi} - E_{midgap} = \frac{3}{4} kT * \ln\left(\frac{m_p^*}{m_n^*}\right)$

$$E_{Fi} - E_{midgap} = \frac{3}{4} * 8.617 * 10^{-5} * 300 * \ln\left(\frac{0.48}{0.067}\right) = 0.038177eV$$

- d. Bandgap data: Si(eV) = 1.12; GaAs(eV)=1.42; Ge(eV)=0.66

Part A

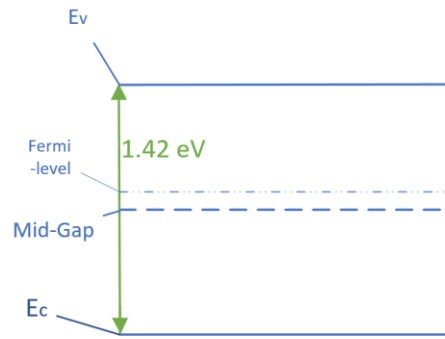


Part B



Student's Name: Jacob Vaught

Part C



Student's Name: Jacob Vaught

Problem 5: (3+3 = 6 points)

(a) Determine the values of n_0 and p_0 in GaAs at $T = 300$ K if $E_F - E_v = 0.25$ eV.

(b) Assuming the value of p_0 in part (a) remains constant, determine the values of $E_F - E_v$ and n_0 at $T = 400$ K. Use N_c and N_v and $E_c = E_g$ from the Room Temperature provided at the end of the questions.

a) GaAs $E_g = 1.42$ eV

$$E_c - E_F = 1.42 - 0.25 = 1.17 \text{ eV}$$

$$n_0 = N_c * \exp\left(-\frac{E_c - E_F}{kT}\right) \rightarrow n_0 = 4.7 * 10^{17} * e^{\left(-\frac{1.17}{8.617 * 10^{-5} * 300}\right)} = 1.038 * 10^{-2}$$

$$p_0 = N_v * \exp\left(-\frac{E_F - E_v}{kT}\right) = 7.0 * 10^{18} * e^{\left(-\frac{0.25}{8.617 * 10^{-5} * 300}\right)} = 4.417 * 10^{14}$$

b) $p_0 = 4.417 * 10^{14}$

$$E_F - E_v = kT * \ln\left(\frac{N_v}{p_0}\right) = 8.617 * 10^{-5} * 400 * \ln\left(\frac{7.0 * 10^{18}}{4.417 * 10^{14}}\right) = 0.33333 \text{ eV}$$

$$n_0 = 4.7 * 10^{17} * e^{\left(-\frac{1.42 - 0.3333}{8.617 * 10^{-5} * 400}\right)} = 9544.4508$$

Student's Name: Jacob Vaught

Problem 6:(2+4 =6 points)

Assume that silicon(Si), germanium(Ge), and gallium arsenide(GaAs) each have dopant concentrations of $N_d = 1 \times 10^{13} \text{ cm}^{-3}$ and $N_a = 2.5 \times 10^{13} \text{ cm}^{-3}$ at $T = 300 \text{ K}$. For each of the three materials:

(a) Is this material n type or p type?

(b) Calculate n_0 and p_0 .

Use value of n_i the from the table at the end of the questions.

n_i : $Si = 1.5 \times 10^{10}$; $Ge = 2.4 \times 10^{13}$; $GaAs = 1.8 \times 10^6$

a) Because $N_a > N_d$, all 3 materials are P-types.

b) We can nullify N_d because N_a is larger, thus $N_d = 0$, and $N_a = N_a - N_d = 1.5 \times 10^{13}$

For P-type material: $p_0 \cong N_a \cong 1.5 \times 10^{13}$

Silicon

$$n_0 = \frac{n_i^2}{p_0} = \frac{(1.5 \times 10^{10})^2}{1.5 \times 10^{13}} = 1.5 \times 10^7$$

Germanium (because n_i is on the same magnitude as $N_a - N_d$)

$$n_0 = \frac{(N_a - N_d)}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2} = \frac{(1.5 \times 10^{13})}{2} + \sqrt{\left(\frac{1.5 \times 10^{13}}{2}\right)^2 + (2.4 \times 10^{13})^2}$$
$$\cong 3.265 \times 10^{13}$$

Gallium Arsenide

$$n_0 = \frac{n_i^2}{p_0} = \frac{(1.8 \times 10^6)^2}{1.5 \times 10^{13}} = 0.216$$

Student's Name: Jacob Vaught

Problem 7:(2+2+2+2 =8points)

A hypothetical semiconductor has an intrinsic carrier concentration(n_i) of $1.0 \times 10^{10}/\text{cm}^3$ at 300K, it has conduction and valence band effective densities of states, N_c and N_v , both equal to $10^{19}/\text{cm}^3$.

(a) What is the bandgap, E_g ?

(b) If the semiconductor is doped with $N_d = 1 \times 10^{16}$ donors/ cm^3 , what are the equilibrium electron and hole concentrations at 300K?

(c) If the same piece of semiconductor, already having $N_d = 1 \times 10^{16}$ donors/ cm^3 , is also doped with $N_a = 2 \times 10^{16}$ acceptors/ cm^3 , what are the new equilibrium electron and hole concentrations at 300K?

(d) Consistent with your answer to Part (c), what is the Fermi level position with respect to the intrinsic Fermi level, $E_F - E_i$?

a.)

$$n_i^2 = N_c * N_v * e^{\left(\frac{-E_g}{kT}\right)} \rightarrow E_g = -kT * \ln\left(\frac{n_i^2}{N_c N_v}\right) \rightarrow E_g = -8.617 * 10^{-5} * 300 * \ln\left(\frac{(10^{10})^2}{(10^{19})^2}\right) \\ = 1.0714\text{eV}$$

b.)

(Material is N-TYPE) $n = 10^{16}$; $p = \frac{(10^{10})^2}{10^{16}} = 10^4$

c.)

(Material is P-TYPE) $p = 2 * 10^{16} - 1 * 10^{16} = 10^{16}$; $n = \frac{(10^{10})^2}{10^{16}} = 10^4$

d.)

$$E_i - E_F = kT * \ln\left(\frac{N_A - N_d}{n_i}\right) \rightarrow 8.617 * 10^{-5} * 300 * \ln\left(\frac{10^{16}}{10^{10}}\right) = 0.357\text{eV}$$

Student's Name: Jacob Vaught

Table B.4 | Silicon, gallium arsenide, and germanium properties ($T = 300$ K)

Property	Si	GaAs	Ge
Atoms (cm^{-3})	5.0×10^{22}	4.42×10^{22}	4.42×10^{22}
Atomic weight	28.09	144.63	72.60
Crystal structure	Diamond	Zincblende	Diamond
Density (g/cm^3)	2.33	5.32	5.33
Lattice constant ($\text{\AA}$)	5.43	5.65	5.65
Melting point ($^{\circ}\text{C}$)	1415	1238	937
Dielectric constant	11.7	13.1	16.0
Bandgap energy (eV)	1.12	1.42	0.66
Electron affinity, χ (V)	4.01	4.07	4.13
Effective density of states in conduction band, N_c (cm^{-3})	2.8×10^{19}	4.7×10^{17}	1.04×10^{19}
Effective density of states in valence band, N_v (cm^{-3})	1.04×10^{19}	7.0×10^{18}	6.0×10^{18}
Intrinsic carrier concentration (cm^{-3})	1.5×10^{10}	1.8×10^6	2.4×10^{13}
Mobility ($\text{cm}^2/\text{V-s}$)			
Electron, μ_n	1350	8500	3900
Hole, μ_p	480	400	1900
Effective mass ($\frac{m^*}{m_0}$)			
Electrons	$m_l^* = 0.98$	0.067	1.64
	$m_t^* = 0.19$		0.082
Holes	$m_{lh}^* = 0.16$	0.082	0.044
	$m_{hh}^* = 0.49$	0.45	0.28
Density of states effective mass			
Electrons ($\frac{m_{dn}^*}{m_0}$)	1.08	0.067	0.55
Holes ($\frac{m_{dp}^*}{m_0}$)	0.56	0.48	0.37
Conductivity effective mass			
Electrons ($\frac{m_{cn}^*}{m_0}$)	0.26	0.067	0.12
Holes ($\frac{m_{cp}^*}{m_0}$)	0.37	0.34	0.21